Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Applied ML, Generative AI, Anomaly Detection, Time-Series | Production ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

nishal.silva@hotmail.com | +1 613 200 2030 | Toronto, ON | nishal.xyz | linkedin.com/in/nishal-silva

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18th October 2026

Re: Senior Data Scientist (Gen AI, Machine Learning) – Extreme Networks

Dear Hiring Manager,

I am applying for the Senior Data Scientist (Gen AI, Machine Learning) role at Extreme Networks. I am a Machine Learning Research Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production under hard latency limits. Your AI Core group's focus on real-time multi-agent systems for network management, with anomaly detection and data mining at the centre, is close to the pattern and anomaly detection on streaming data that I have done for most of my career.

Three parts of my work match your posting. For production ownership, I led ML and computer vision systems at MAS Holdings from design to deployment, including care-label inspection that cut inspection time by 99.5% and raised operational efficiency by up to 300%, with real-time IoT ingestion and ETL pipelines across distributed sites. For real-time ML, my published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline at F1 0.76 (Journal of the Audio Engineering Society, 2026), and I owned its lifecycle from data acquisition to deployment and monitoring on AWS with Docker and Airflow. For generative AI, I have worked with diffusion and VAE based synthetic data generation, and I am currently fine-tuning a pre-trained audio foundation model with LoRA. I also publish regularly, with 10+ peer-reviewed papers, four open datasets on Zenodo and an open-source C++17 library.

I should be direct about the gaps. My generative AI work is applied and tool-level, not three years of production RAG or agent systems, and I have not used GNNs or GraphRAG in production. I have also worked on AWS rather than Azure or GCP, and not on Spark, Flink or Kafka. I have limited experience leading a research organisation, although I mentored engineers and set up code review and CI/CD practices at MAS. I learn new stacks quickly and would aim to contribute first on anomaly detection and evaluation design.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience can support the AI Core group at Extreme Networks. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD